



Cybersecurity Ethics

Name: Isaac Adebayo

Class: MSDS 575

Date: February 7th, 2026

What is Cybersecurity?



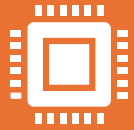
Cybersecurity involves protecting computer system, network, data transfer, hardware and software from unauthorized users irrespective of the intention or nature.



Some common areas cybersecurity applies to are:

- Network security
- Application security
- Information/Data security
- Cloud security
- Operational security

Layer of Cybersecurity



SOFTWARE: Cybersecurity software protects system network and data from unauthorized activities examples include anti-virus, firewall, VPN, network analyzers such as Wireshark and port scanner, anti-keylogger software for keyboard, file and cloud encryption software like ax crypt



HARDWARE: Protect the integrity of computer hardware from unauthorized users , examples include physical firewall, endpoint encryption for hardware such as YubiKey for multifactor authentication prior to login, and tamper proofing hardware, faraday cage or faraday bag to prevent EMS pulse attacks etc ...

Encryption



What is Encryption?



Encryption is securing data via standard algorithm to protect information via public transfer (a subset of cybersecurity) with the aim of ensuring privacy and integrity of system, network and communication during data transfer or data storage

OSI Layer Encryption

OSI layer model of encryption

- Layer 1 – Physical
- Layer 2 – Data Link
- Layer 3 – Network
- Layer 4 – Transport
- Layer 5 – Session
- Layer 6 – Presentation
- Layer 7 - Application

Example of encryption for each layer model

- Layer 1 – Physical (File vault or disk encryption encrypt hardware)
- Layer 2 – Data Link (WPA3, link layer encryption to secure wireless link, P25 or TETRA radio encryption to secure voice and data transmission between radio)
- Layer 3 – Network (IP sec to encrypt data packets, VPN to mask IP address)
- Layer 4 – Transport (SSL for transport session)
- Layer 5 – Session (end to end encryption between sessions)
- Layer 6 – Presentation (TLS for data encryption and decryption across the web)
- Layer 7 – Application (PGP for email using AES)



Example of Cybersecurity Breach



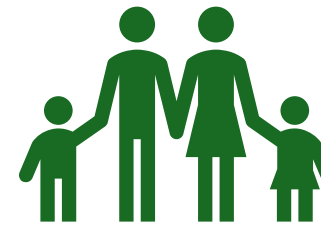
Almost all cyberbreach involve malicious collection of data (personal or financial) or denial of service/ access. The data science field is involved in the analysis of data, some common cybersecurity breach over the past 10 years are:

Equifax (2017)

Yahoo (2013-2014)

LinkedIn (2019/2021)

Facebook (2019)



Over 90% of breaches are caused by human error.

Screenshot of Cybersecurity Paper

Cybersecurity: Past, Present and Future

By

Shahid Alam, PhD

Department of Computer Engineering
Adana Alparslan Turkes Science and Technology University
Adana, Turkey
salam@atu.edu.tr

© Shahid Alam – 2 October 2022

Link: <https://arxiv.org/pdf/2207.01227>

